

Curated Research Articles

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- **Overcoming Interphase-Dependent Li⁺ Transport via Tridentate Coordination Engineered Solvation Sheath** — score: 1.000 The article discusses a novel approach to enhance the performance of gel polymer electrolytes in lithium metal batteries by engineering the coordination of TFSI[−] from monodentate to tridentate. This modification leads to a more ion-conductive solid-electrolyte interphase (SEI), minimizing resistance and improving battery longevity, with impressive cycling stability demonstrated in various cell configurations. This research suggests that the regulation of ligand denticity can serve as a crucial design principle for future electrolyte development. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Regulating Multi-Metal Ion Orbital Interactions to Enhance Kinetics and Structural Stability in Lithium Manganese Iron Phosphate Cathodes** — score: 1.000 The study introduces a novel multi-metal ion orbital engineering approach for enhancing the performance of lithium manganese iron phosphate (LMFP) cathodes, specifically LiMn_{0.475}Fe_{0.475}Ni_{0.03}Cu_{0.01}Ti_{0.01}PO₄. By optimizing electronic distribution and reducing Jahn-Teller distortion through tailored orbital hybridization, the cathode exhibits significantly improved conductivity, structural stability, and impressive cycling performance with 92.3% capacity retention after 1000 cycles at 1C. Journal: *Wiley-Online-Library: Advanced Energy Materials: Table of Contents*
- **An Octopus-Inspired Gel Electrolyte Regulates Cation Migration in Lithium–Sulfur Batteries via Constructing Differentiated Solvation Environments** — score: 1.000 This study presents an innovative gel electrolyte inspired by octopus biology, designed to manage cation migration in lithium-sulfur (Li–S) batteries. The synthesized multi-arm amphiphilic polymer gel selectively inhibits the shuttling of lithium polysulfides while enhancing lithium ion dissociation and transport, leading to impressive cycling stability and capacity retention over extended use. Journal: *Wiley-Online-Library: Advanced Energy Materials: Table of Contents*
- **Quantitative Analysis of Topographic Crosstalk in DART-ESM Arising from Feedback-Loop-Delay-Induced Contact Stiffness Variations in Battery Materials** — score: 1.000 The study investigates topography-induced artifacts complicating the interpretation of signals in Dual AC Resonance Tracking Electrochemical Strain Microscopy (DART-ESM). By analyzing contact stiffness variations using artificial trench structures on silicon as a reference, it quantifies the impact of these artifacts and demonstrates that Cooling Cross-Section Polishing (CCP) significantly mitigates them, thereby enhancing the accuracy of nanoscale electrochemical assessments in battery materials. Journal: *Wiley-Online-Library: Small Methods: Table of Contents*
- **Porous Hybrid Organic–Metal Chalcogenides with Tubular SBUs and Doping-Enhanced Conductivity for Electrochemical Energy Storage** — score: 0.800 This study presents the synthesis of a porous hybrid organic-metal chalcogenide, Cu-I-TMB, which achieves a threefold increase in conductivity through doping with iodine. The material maintains its structure while demonstrating a reversible capacity of 236 mAh g^{−1} in aqueous zinc-iodine batteries, highlighting the potential for improved electrochemical energy storage through structural enhancement and doping strategies. Journal: *Wiley: Small Methods: Table of Contents*
- **Regulating Li-Ions Conduction of Eco-Friendly Cellulose-Based Composite Electrolytes via Amino-Siloxane Functional Grafting for Sustainable Li-Metal Batteries** — score: 0.800 The article discusses a novel approach to enhance Li-ion conductivity in cellulose-based composite electrolytes by employing amino-siloxane functional grafting and sol-gel self-assembly. This method not only improves ionic conductivity and stabilizes the interface for Li-metal batteries, leading to extended cycling life, but also demonstrates significant sustainability benefits in terms of energy conservation and carbon reduction, marking a promising advancement in eco-friendly battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Direct electrode-to-electrode regeneration of end-of-life batteries via electrode–electrolyte interphase dissolution** — score: 0.800 This article discusses a novel method for regenerating spent

lithium-ion battery electrodes by selectively dissolving the electrode–electrolyte interphase (EEI). This approach allows for a remarkable recovery of up to 95% of the original capacity, offering a promising solution for extending the life of batteries. Journal: *RSC - Energy Environ. Sci. latest articles*

- **Unlocking Superionic Conduction by Modulating Electrostatic Interactions in a Zirconium-Based Trigonal Halide Solid Electrolyte** — score: 0.800 This article discusses advancements in zirconium-based halide solid electrolytes, specifically Li_2ZrCl_6 , which exhibit enhanced superionic conductivity through the modulation of electrostatic interactions. The findings suggest that these materials have significant potential for application in solid-state batteries due to their combination of high ionic conductivity and favorable characteristics. Journal: *RSC - EES Batteries latest articles*
- **A kinetic descriptor of solvent polarization for rational electrolyte design in lithium–metal batteries** — score: 0.800 This article introduces a new kinetic descriptor for solvent polarization that integrates bulk transport properties and interfacial stability, providing a framework for the rational design of electrolytes in lithium-metal batteries. The approach aims to enhance battery efficiency by optimizing electrolyte characteristics. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Interfacial ion depletion at high concentration guides electrolyte design for aqueous aluminum-iodine batteries** — score: 0.800 The study by Yang et al. reveals that high concentrations of salt in aqueous multivalent batteries can hinder the effectiveness of functional cations, as they become trapped by aluminum complexes and cannot reach the electrode interface. Identifying an optimal concentration range allows for the development of a more efficient and durable aluminum-iodine battery, providing insights for future electrolyte designs. Journal: *Joule*
- **Plasma Enabled Hierarchical Surface Reconstruction of Nanoengineered, Dendrite-Free Zn Metal for Durable Aqueous Zinc Ion Battery** — score: 0.800 The article discusses a novel approach using low-temperature Argon plasma to modify zinc metal anodes in aqueous zinc-ion batteries, enhancing their stability by reducing surface impurities and roughness. This technique promotes the formation of uniform nanoisland seed layers, which effectively suppress dendrite growth and improve long-term performance compared to untreated zinc anodes, addressing key stability challenges in battery technology. Journal: *Wiley: Small Methods: Table of Contents*
- **Tunnel-Framework Interphase Driving Rapid Zn^{2+} Diffusion for Long-Life Zn Metal Anodes** — score: 0.800 The article presents a novel 3D tunnel-framework interphase made from a Si_3N_4 layer, which enhances Zn^{2+} diffusion and significantly reduces harmful reactions and dendrite formation in aqueous zinc-ion batteries (AZIBs). This engineered interface allows for stable zinc deposition and nucleation, resulting in extended lifespan and high coulombic efficiency of the Zn anodes, demonstrating promising advantages for the commercialization of AZIBs in large-scale energy storage applications. Journal: *Wiley-Online-Library: Advanced Energy Materials: Table of Contents*
- **A Lignin-Targeted Pre-Oxidation Strategy for Mitigating Closed Pore Collapse Toward High-Stability Wood-Based Hard Carbon Anodes for Na-Ion Batteries** — score: 0.800 The study presents a lignin-targeted pre-oxidation method for improving the structural integrity of wood-based hard carbon anodes in sodium-ion batteries. By leveraging the different thermal properties of lignin and cellulose, the process reduces closed-pore collapse during carbonization and optimizes pore architecture, resulting in high reversible capacity and excellent cycle stability across various biomass sources. Journal: *Wiley-Online-Library: Advanced Energy Materials: Table of Contents*
- **Recent Advances of Fiber-Shaped Aqueous Zinc Ion Batteries: From Structure, Principles, Issues to Strategies** — score: 0.700 The article reviews the advancements in fiber-shaped aqueous zinc-ion batteries (FAZIBs), emphasizing their potential for wearable electronics due to their safety and flexibility. It analyzes critical challenges such as zinc dendrite formation and cathode dissolution while proposing strategies for performance enhancement and durability. The review also outlines future directions for optimizing FAZIB designs to improve energy density and stability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Regulation of the Charge Distribution in the Confined Channels of Anion Exchange**

- Membranes Enables Efficient Ion Transport** — score: 0.600 The article presents a novel approach to enhancing ion transport in anion exchange membranes by strategically positioning charges within the membrane’s sub-nanometer channels, rather than on the walls. This design minimizes excessive water uptake, improving the hydrogen-bond network to mimic bulk water, resulting in higher self-diffusion coefficients and achieving optimal conductivity and energy efficiency in electrochemical applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **CVD-grown tunable carbon films for high-performance sodium storage** — score: 0.600 The article discusses the development of tunable carbon films through chemical vapor deposition aimed at enhancing the performance of sodium-ion batteries. These advancements could significantly improve the energy density by utilizing electrode materials with high reversible capacity, positioning sodium-ion technology as a competitive and sustainable energy-storage solution. Journal: *RSC - Energy Environ. Sci. latest articles*
 - **Non-Flammable Li-Diamond Composite Anode Enabling High-Safety Lithium Metal Batteries** — score: 0.500 The study presents a novel ultrathin Li-diamond composite anode that significantly improves the safety and electrochemical performance of lithium metal batteries. The diamond modification not only prevents combustion during thermal abuse by isolating lithium from oxygen but also enables the development of intrinsically safe lithium anodes when used with non-flammable electrolytes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **Hydrogels as interfacial materials for the water-energy nexus** — score: 0.400 The article reviews the use of hydrogels as versatile materials in managing crucial transport processes at interfaces related to the water-energy nexus. It highlights how the chemistry and design of hydrogels can enhance efficiencies for technologies such as solar water generation, atmospheric water harvesting, and hydrovoltaic power generation, ultimately contributing to sustainable solutions in water and energy management. Journal: *Joule*
 - **Defect-Directed Regeneration of LiCoO₂ Cathodes via Gradient Cobalt Reprogramming** — score: 0.400 The article discusses a novel approach to regenerate LiCoO₂ cathodes in lithium-ion batteries by implementing gradient cobalt reprogramming. This method not only aims to enhance defect homogenization but also emphasizes the importance of optimizing material properties rather than solely restoring the original structure, paving the way for more effective and sustainable battery recycling practices. Journal: *RSC - Energy Environ. Sci. latest articles*
 - **Discharge-Triggered Electronic Delocalization via Asymmetric Motifs Suppresses Jahn-Teller Distortion in Mn-Based Layered Oxides** — score: 0.400 The study examines Mn-based layered oxide cathodes for sodium-ion batteries, focusing on how discharge-triggered electronic delocalization can mitigate Jahn-Teller distortion caused by the formation of Mn³⁺. The findings suggest that implementing asymmetric motifs in these materials enhances their structural stability and overall performance. Journal: *RSC - Energy Environ. Sci. latest articles*
 - **Comparing the safety of graphite and silicon negative active materials in lithium ion batteries** — score: 0.400 The study evaluates the safety of graphite and silicon as negative active materials in lithium-ion batteries, highlighting that while silicon is an attractive option for enhancing energy density, it poses significant risks due to exothermic reactions associated with structural degradation and gas evolution during use. The findings suggest that these safety concerns necessitate further investigation into silicon’s practical application in battery technology. Journal: *RSC - EES Batteries latest articles*
 - **Thermodynamic anion anchoring in polyurethane electrolytes for fast-charging lithium metal batteries** — score: 0.400 The article discusses the development of polyurethane electrolytes designed with thermodynamic anion anchoring, which enhances the performance of lithium metal batteries, particularly during fast-charging processes. This innovation aims to improve the stability and efficiency of battery systems, paving the way for advancements in energy storage technologies. Journal: *ScienceDirect Publication: Nano Energy*
 - **AI-Driven Approaches for Navigating Battery Complexity** — score: 0.400 This mini-review

discusses the increasing role of artificial intelligence in simplifying the complexities associated with battery materials and electrochemical systems. It highlights recent advancements in AI applications within battery research, emphasizing how these approaches can enhance understanding and development in the field. Journal: *RSC - EES Batteries latest articles*

- **Design-Driven Multi-Element Doping for Long-Life High-Ni NCM Cathodes: Criteria-Guided Dopant Selection and One-Pot Element-Specific Intraparticle Placement** — score: 0.400 The article explores a novel approach to enhancing the longevity of high-nickel NCM cathodes through multi-element doping. It details a criteria-guided method for selecting dopants and employs a one-pot process for the targeted placement of these elements within the particles, aiming to improve performance and lifespan in energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Early Warning of Battery Thermal Runaway Enabled by Ag-Decorated SnO₂ and Machine Learning: Diethyl Carbonate Recognition** — score: 0.300 The study highlights the development of a sensor using silver-decorated SnO₂ for rapid detection of diethyl carbonate (DEC) in lithium-ion batteries, showcasing a 12.6-fold improvement in response compared to unmodified SnO₂. Coupled with machine learning techniques, the sensor achieved a remarkable 99.3% classification accuracy for identifying carbonates, offering promising applications for early warning systems to prevent battery thermal runaway and enhance safety. Journal: *Wiley: Small Methods: Table of Contents*
- **Molecularly engineered dual-network eutectogel electrolytes for high-loading and stable zinc-iodine batteries** — score: 0.300 The article discusses advancements in aqueous zinc-iodine batteries (AZIBs) through the development of engineered dual-network eutectogel electrolytes, which enhance the stability of zinc anodes and iodine cathodes. This innovation aims to address the current limitations of AZIBs, promoting their viability for safe and cost-effective energy storage solutions. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Unlocking High-Performance Aqueous Zn-S Batteries via Synergistic Supramolecular-Redox Mediator Electrolyte Engineering** — score: 0.300 The article discusses advancements in the performance of aqueous zinc-sulfur (Zn-S) batteries by developing a synergistic supramolecular-redox mediator electrolyte. This innovative approach enhances the battery's efficiency and longevity, paving the way for high-performance energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Synergistically homogenizing and accelerating Li⁺ transport via nanoporous composite separator for high stable lithium metal batteries** — score: 0.300 The article discusses a novel nanoporous composite separator designed to enhance the uniformity and speed of lithium ion transport in lithium metal batteries. By optimizing the separator's structure, the researchers aim to improve the batteries' stability and overall performance, addressing common challenges in lithium metal technology. Journal: *ScienceDirect Publication: Nano Energy*
- **Real-Space Molecular Resolution Imaging of Graphite-Anchored MOF Surfaces via Atomic Force Microscopy at the Liquid-Solid Interface** — score: 0.300 This article discusses a novel method for preparing metal-organic frameworks (MOFs) anchored to highly oriented pyrolytic graphite (HOPG) by covalent grafting, allowing for advanced molecular-resolution imaging using liquid atomic force microscopy (AFM). The technique preserves the integrity of MOF surfaces, enabling detailed characterization of local structural features across different MOF types, demonstrating the effectiveness of this approach for real-time monitoring of surface dynamics. Journal: *Wiley-Online-Library: Small Methods: Table of Contents*
- **Precision Atomic Doping of Nanoalloy Electrocatalysts: Unlocking New Paths for Efficient Hydrogen Evolution Reaction** — score: 0.200 This article review examines the role of precision atomic doping in enhancing the performance of nanoalloy electrocatalysts for the hydrogen evolution reaction, addressing the challenges of efficiency and stability. It provides insights into the mechanisms of doping, categorizes various dopants, and introduces a metric for assessing performance improvements, while also discussing future directions for research in hydrogen production technologies.

- **Unraveling Multiscale Interfacial Dynamics for High-Flux Hydrogen Peroxide Electrosynthesis** — score: 0.200 This review presents a comprehensive multiscale framework for electrochemical hydrogen peroxide synthesis, integrating molecular dynamics, mesoscale transport mechanics, and macroscopic system stability. It emphasizes the interconnectedness of various scales in optimizing selectivity and durability of H₂O₂ production under industrial conditions, and highlights potential advancements in characterization and design for enhanced scalability and performance. Journal: *Wiley-Online-Library: Advanced Energy Materials: Table of Contents*
- **Strategies for Mitigating the Dissolution of Organic Electrodes in Rechargeable Batteries** — score: 0.200 The article discusses various strategies aimed at reducing the dissolution of organic electrodes in rechargeable batteries, which is a significant challenge affecting battery performance and lifespan. The authors propose innovative solutions and materials that could enhance the stability of these electrodes, contributing to more efficient energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **A solar-driven ultra-long cycling Li-CO₂ battery enabled by Si-supported bifunctional cathode** — score: 0.200 The article discusses the development of an innovative lithium-carbon dioxide battery that utilizes solar energy for prolonged cycling performance. This battery features a silicon-supported bifunctional cathode, aiming to enhance efficiency and sustainability in energy storage solutions. Journal: *ScienceDirect Publication: Nano Energy*
- **A Synergistically Integrated Spinel High-Entropy oxide@graphitic Carbon Nitride Nanocomposite Synthesized via a Facile Route for a High-Performance Supercapacitor Electrode** — score: 0.200 The article presents the development of a high-performance supercapacitor electrode by combining spinel high-entropy oxides with graphitic carbon nitride in a novel nanocomposite. This synergistic approach enhances conductivity and charge transfer, yielding an impressive specific capacitance of 3263 F/g, while also improving structural stability and durability during charge-discharge cycles, setting a new standard for energy storage materials. Journal: *Wiley-Online-Library: Small Methods: Table of Contents*
- **Battery minerals for neural inhibition** — score: 0.200 The article discusses the innovative use of triphylite (LiFePO₄) as a lithium-containing battery cathode within a bioelectronic device. This setup enables precise and localized delivery of lithium ions to neurons, offering a new approach for neural inhibition in biomedical applications. Journal: *Nature Materials*
- **Stabilizing Ultra-High-Nickel Oxide Cathodes via Residual Strain Control** — score: 0.100 The article discusses a novel approach to enhance the stability of ultra-high-nickel oxide cathodes for energy storage applications by managing residual strain. The authors explore how effective strain control can lead to improved performance and longevity of these cathodes, addressing challenges associated with their use in energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Soft-Chemistry Routes to Entropy-Engineered Nanomaterials: Complexity-Driven Performance in Redox Electrocatalysis** — score: 0.100 The article reviews the innovative use of soft-chemistry methods to develop high-entropy nanomaterials (HENMs) that enhance redox electrocatalysis through their complex compositions. These nanomaterials showcase improved performance by creating synergistic active sites and tailored electronic properties, leading to greater durability compared to traditional electrocatalysts. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Tuning Interfacial Water Dynamics via Gd-Doped Cu₂O for High-Rate Selective CO₂-to-Ethanol Conversion** — score: 0.000 The study investigates a Gd-doped Cu₂O electrocatalyst that alters interfacial water dynamics to enhance the electrochemical conversion of CO₂ to ethanol. By disrupting rigid hydrogen-bonding networks, the catalyst creates a free-water-rich environment that improves proton-coupled electron transfer, achieving high Faradaic efficiencies and ethanol selectivity at relevant current densities, while also suppressing competing pathways. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Phosphate-Buffered Synthesis of SnO₂ Nanoparticles With Interfacial Proton Buffering for High-Performance and Stable Perovskite Solar Cells** — score: 0.000 The study introduces a phosphate-buffered synthesis method for SnO₂ nanoparticles, which optimizes proton buffering at the interface with perovskite layers. This innovation improves film quality, enhances charge extraction, and leads to a peak power conversion efficiency of 26.1% in solar cells, while also significantly increasing operational stability by retaining over 85% efficiency after 1000 hours of exposure to light, making it a scalable solution for high-performance perovskite solar cells. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Modulating Interlayer Chemical Bonding Heterogeneity for Electron-Phonon Decoupling: Unlocking High Thermoelectric Performance in Layered CuCrTi₂Se₆** — score: 0.000 The study presents a strategy to enhance the thermoelectric performance of layered CuCrTi₂Se₆ by substituting chromium for copper, effectively modulating interlayer chemical bonding. This approach achieves electron-phonon decoupling, resulting in improved carrier mobility and decreased lattice thermal conductivity, culminating in a record thermoelectric figure of merit ($ZT \approx 1.0$) and efficient device performance at high temperatures. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **GPR15-guided CD8⁺ T regulatory cells control intestinal inflammation** — score: 0.000 The study identifies the role of GPR15 in guiding CD8⁺ T regulatory cells, which play a crucial part in managing intestinal inflammation. This research highlights the therapeutic potential of targeting these cells to mitigate inflammatory bowel diseases. Journal: *Nature*
- **Distributed control circuits across a brain-and-cord connectome** — score: 0.000 The article discusses the architecture and functionality of distributed control circuits in the neural connectome linking the brain and spinal cord. The findings shed light on how these circuits coordinate complex behaviors and motor functions, highlighting their significance in understanding neural processes and potential implications for treating disorders related to motor control. Journal: *Nature*
- **Targeting Cancer-Specific Mutations with RNA-Triggered Chromatin Shredding** — score: 0.000 The article discusses a novel therapeutic approach that utilizes RNA-triggered mechanisms to selectively shatter chromatin associated with cancer-specific mutations. This strategy aims to enhance the precision of cancer treatment by directly targeting the genetic alterations that drive cancer progression, potentially leading to more effective outcomes in oncology. Journal: *Nature*
- **Fifty years since a simple equation described the chaos of biology** — score: 0.000 The article reflects on the impact of chaos theory over the past fifty years in understanding biological population dynamics, highlighting how complex and unpredictable biological systems can be effectively described using straightforward mathematical equations. This highlights the significance of mathematical modeling in capturing the chaotic nature of biological interactions. Journal: *Nature*
- **How AI is reshaping discovery in maths and physics** — score: 0.000 The article discusses the transformative influence of artificial intelligence on mathematical and physical research, highlighting its role in reshaping inquiry and comprehension rather than substituting human creativity. By redefining the way questions are formulated and explored, AI is enhancing the discovery process in these scientific domains. Journal: *Nature*
- **Why are so many young people getting cancer? What researchers do and don't know** — score: 0.000 The article discusses the rise in cancer cases among young people, highlighting key factors that researchers are identifying as potential contributors, which may differ across various tumor types. Despite ongoing investigations, uncertainties remain regarding the exact causes of this worrying trend. Journal: *Nature*
- **Bots are scraping open data — how should researchers respond?** — score: 0.000 The article discusses concerns among researchers regarding the increasing capabilities of artificial intelligence to scrape open data sets, raising alarms about potential loss of control over research information. It emphasizes the need for scientists to respond strategically to protect their data in the face of growing AI applications. Journal: *Nature*

- **AI is taking on antibiotic resistance — here's how** — score: 0.000 The article discusses the role of advanced artificial intelligence tools in accelerating the discovery of novel antibiotics to combat antibiotic resistance. By leveraging AI technology, researchers aim to enhance the efficiency and effectiveness of developing solutions to this growing public health challenge. Journal: *Nature*
- **Nuclear-fusion firm says plant will deliver electricity to grid – but big questions remain** — score: 0.000 Commonwealth Fusion Systems has announced plans for its ARC fusion power plant, claiming it will be capable of delivering electricity to the grid. However, the announcement raises significant questions regarding the feasibility and operational readiness of fusion energy technologies. Journal: *Nature*
- **Sustainability or dystopia? What past patterns tell us about where society is heading** — score: 0.000 The article discusses a new book that explores historical societal transformations through the lens of environmental tipping points, aiming to provide insights into potential future scenarios for society. It evaluates whether current trends may lead to sustainable outcomes or a dystopian future. Journal: *Nature*